

COMPARATIVE SEQUENCE ANALYSIS OF TEM-TYPE BETA-LACTAMASE GENES IN BACTERIAL ISOLATES

ABSTRACT

Beta-lactam antibiotics are widely used to treat bacterial infections. However, the production of beta-lactamase enzymes is an important mechanism through which bacteria develop resistance to these antibiotics. TEM-type beta-lactamases are encoded by blaTEM genes and occur in different sequence variants.

The present study aimed to compare five nucleotide sequences representing TEM-type beta-lactamase genes using bioinformatics tools. The selected sequences were KJ923009.1, KJ923008.1, NG_050151.1, NG_050166.1 and KJ923002.1. Multiple sequence alignment was performed using Clustal Omega, followed by pairwise sequence identity analysis using Python and Biopython. A phylogenetic tree was constructed to visualize the relationships among the sequences. Variable nucleotide positions were identified, and preliminary protein translation and amino acid comparison were performed.

The initial analysis reported an alignment length of 861 nucleotides, eight variable nucleotide positions and five variable amino acid positions. Each sequence produced a preliminary translation of 287 amino acids. These findings provide an initial comparison of sequence variation among the selected genes. Further validation of the alignment, sequence identity calculations and coding frame is required before the numerical results can be considered final.

Keywords: Bioinformatics, TEM-type beta-lactamase, antibiotic resistance, sequence alignment, pairwise identity, phylogenetic analysis, nucleotide variation.

INTRODUCTION

1. Background

Antibiotic resistance is a major challenge in the treatment of bacterial infections. It occurs when bacteria develop mechanisms that reduce or prevent the effectiveness of antimicrobial drugs.

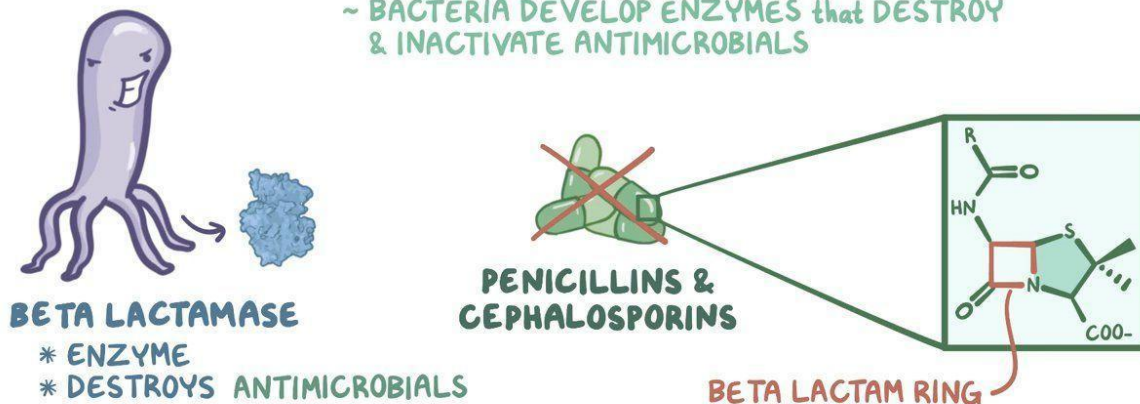
One important mechanism of antibiotic resistance is the production of beta-lactamase enzymes. These enzymes hydrolyze the beta-lactam ring present in many antibiotics, including penicillins and certain cephalosporins, thereby reducing their antibacterial activity.

TEM-type beta-lactamases are enzymes encoded by genes commonly designated blaTEM. Different TEM variants may contain nucleotide substitutions that lead to amino acid changes in the encoded protein. Such changes can influence enzyme properties, although the effect of an individual substitution must be experimentally established or supported by appropriate functional evidence.

MECHANISMS of ANTIMICROBIAL RESISTANCE

1. ANTIBIOTIC INACTIVATION or MODIFICATION

~ BACTERIA DEVELOP ENZYMES that DESTROY & INACTIVATE ANTIMICROBIALS



2. Importance of bioinformatics

Bioinformatics combines biological knowledge with computational methods to analyze biological data. DNA sequence analysis helps researchers identify similarities and differences among genes, investigate evolutionary relationships and generate hypotheses about genetic variation.

Several bioinformatics methods are useful for comparing beta-lactamase genes:

- **Multiple sequence alignment:** arranges sequences to identify conserved and variable positions.

- **Pairwise sequence identity:** measures the percentage of matching nucleotides between two sequences.
- **Phylogenetic analysis:** estimates relationships among sequences based on their similarities and differences.
- **Nucleotide variation analysis:** identifies positions where sequences differ.
- **Protein sequence analysis:** investigates whether nucleotide differences correspond to amino acid substitutions.

In this project, these methods were used to conduct a comparative analysis of five selected TEM-type beta-lactamase gene sequences.

3. Rationale of the study

Different TEM variants can share high nucleotide similarity while containing specific sequence differences. Comparing their nucleotide and translated protein sequences can help identify conserved regions, variable positions and patterns of relatedness.

The study provides a small-scale example of how bioinformatics tools can be applied to antibiotic-resistance-associated genes.

AIM

To compare selected TEM-type beta-lactamase gene sequences using bioinformatics tools and investigate their nucleotide variation, pairwise similarity, preliminary protein variation and phylogenetic relationships.

OBJECTIVES

1. To obtain five nucleotide sequences representing TEM-type beta-lactamase genes.
2. To perform multiple sequence alignment using Clustal Omega.
3. To calculate pairwise nucleotide sequence identity.
4. To construct a phylogenetic tree to visualize sequence relationships.
5. To identify variable nucleotide positions in the alignment.
6. To translate the nucleotide sequences into preliminary protein sequences.
7. To identify variable amino acid positions among the translated sequences.
8. To interpret the findings in relation to genetic variation and the limitations of sequence-based analysis.

MATERIALS AND METHODS

1. Materials and tools

The following resources and tools were used or consulted during the analysis.

Resource/tool	Purpose
NCBI Nucleotide database	Retrieval and identification of nucleotide records
Clustal Omega	Multiple sequence alignment
Google Colab	Execution of Python code
Biopython	Sequence handling and translation
Pandas	Tabulation and export of analytical results
NGPhylogeny.fr or the phylogenetic tool actually used	Construction and visualization of the phylogenetic tree
Microsoft Excel	Organization of sequence identity results

2. Selection of sequences

Five nucleotide sequence records were selected for comparative analysis.

Sequence label	Accession or sequence identifier	Reported TEM variant
S1	KJ923009.1	TEM-116
S2	KJ923008.1	TEM-116
S3	NG_050151.1:101–961	TEM-106
S4	NG_050166.1	TEM-122
S5	KJ923002.1	TEM-1

3. Sequence retrieval

The selected nucleotide records were identified using their accession numbers in the NCBI Nucleotide database. The relevant nucleotide sequences were collected in FASTA format for subsequent analysis.

FASTA format stores each sequence under a header beginning with the symbol > followed by its identifier and description.

4. Multiple sequence alignment

The five nucleotide sequences were aligned using Clustal Omega.

Multiple sequence alignment places homologous nucleotide positions in corresponding columns, making it possible to compare conserved and variable regions across sequences.

The resulting alignment was reported to contain 861 columns. The alignment was saved in a Clustal-format file and converted to aligned FASTA format for downstream analysis.

5. Pairwise sequence identity analysis

Pairwise nucleotide identity was calculated using Python and Biopython. Each sequence was compared with every other sequence to determine the proportion of matching nucleotide positions.

The percentage identity is calculated as:

Percentage identity = (Number of matching positions / Number of positions compared) × 100

For the reported 861-position alignment, the denominator must be defined consistently for all comparisons. If gaps or ambiguous nucleotides are present, the treatment of those positions should be specified.

6. Phylogenetic analysis

A phylogenetic tree was constructed from the selected sequences to visualize their estimated relationships.

Sequences that cluster together in a tree share greater similarity under the analysis used than sequences separated by more divergent branches. However, the interpretation depends on the alignment, tree-building method, evolutionary model and branch-support values.

7. Identification of variable nucleotide positions

A Python script was used to examine each alignment column. A column was classified as variable when more than one nucleotide state was present among the five sequences.

The analysis reported eight variable nucleotide positions. These positions were recorded in a CSV file for further inspection.

8. Preliminary protein translation

The aligned nucleotide sequences were translated using Biopython and the standard genetic code. The initial analysis reported 287 amino acids per sequence and no internal stop codons.

Because the sequences were translated directly from the aligned DNA, the coding frame, terminal codon and correspondence to the annotated coding sequences must be verified before interpreting the translated products as complete TEM proteins.

9. Amino acid variation analysis

The preliminary protein sequences were compared position by position. A position was classified as variable when more than one amino acid was observed among the five sequences.

The initial comparison reported five variable amino acid positions. These positions require confirmation after validating the coding frame and protein sequence boundaries.

WORKFLOW DIAGRAM

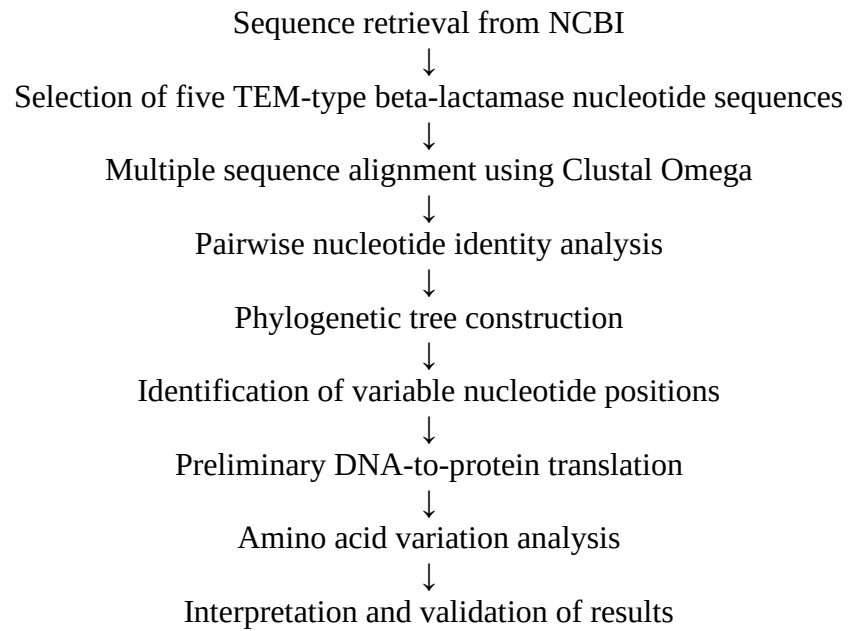


Figure 1. Workflow for comparative sequence analysis of TEM-type beta-lactamase genes.

RESULTS

1. Multiple sequence alignment

Five nucleotide sequences were analyzed using Clustal Omega. The resulting alignment was reported to contain 861 positions.

The alignment provided a common framework for comparing nucleotide positions across the selected sequences. Conserved positions represent nucleotide states shared across sequences, whereas variable positions indicate sequence differences.

Table 1. Summary of multiple sequence alignment

Parameter	Preliminary result
Number of sequences	5
Alignment length	861 positions
Alignment software	Clustal Omega
Alignment format	Clustal, converted to FASTA

2. Pairwise sequence identity

Pairwise identity was calculated for all unique pairs of the five sequences. The initial matrix is shown below, with values rounded to two decimal places.

Table 2. Preliminary pairwise nucleotide identity matrix (%)

Sequence	S1: KJ923009.1	S2: KJ923008.1	S3: NG_050151.1	S4: NG_050166.1	S5: KJ923002.1
S1: KJ923009.1	100.00	100.00	99.19	99.42	99.54
S2: KJ923008.1	100.00	100.00	99.19	99.42	99.54
S3: NG_050151.1	99.19	99.19	100.00	99.54	99.65
S4: NG_050166.1	99.42	99.42	99.54	100.00	99.88
S5: KJ923002.1	99.54	99.54	99.65	99.88	100.00

The preliminary matrix suggests high nucleotide similarity among the selected sequences. However, high sequence identity alone does not establish identical biological function or identical antibiotic-resistance properties.

3. Phylogenetic analysis

A phylogenetic tree was constructed to visualize the relationships among the five selected sequences.

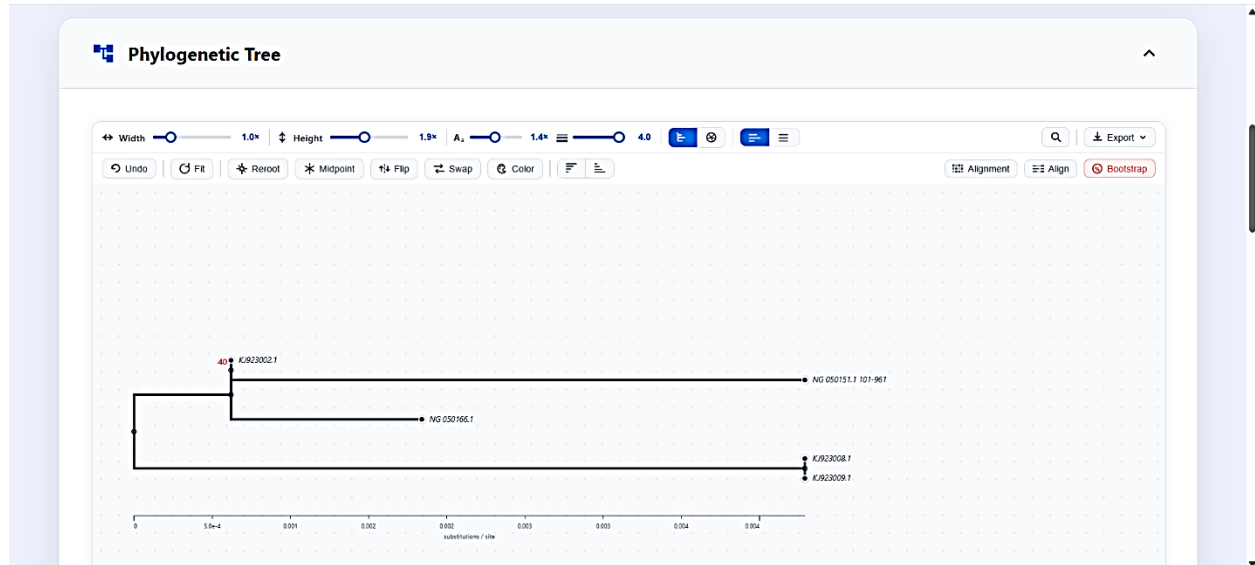


Figure 2. Phylogenetic tree of the five selected TEM-type beta-lactamase gene sequences. The tree displays relationships inferred from the sequence data using the specified phylogenetic method.

4. Variable nucleotide positions

The Python-based comparison reported eight variable nucleotide positions among the five aligned sequences.

Table 3. Summary of nucleotide variation

Parameter	Preliminary result
Number of sequences compared	5
Alignment length	861 positions
Variable nucleotide positions	8

The variable positions represent nucleotide columns in which the selected sequences differ. The complete position-by-position results are available in the generated file blaTEM_variable_positions.csv.

The nucleotide states at each position should be examined before identifying specific substitutions or discussing their possible biological significance.

5. Preliminary protein translation and amino acid variation

The nucleotide sequences were translated using Biopython. The initial analysis reported 287 amino acids per sequence, no internal stop codons and five variable amino acid positions.

Table 4. Preliminary protein comparison

Parameter	Preliminary result
Number of translated sequences	5
Reported translation length	287 amino acids per sequence
Internal stop codons reported	0
Variable amino acid positions	5

DISCUSSION

The present study used bioinformatics methods to compare five selected TEM-type beta-lactamase gene sequences at the nucleotide and preliminary protein levels.

1. Sequence similarity

The preliminary pairwise identity matrix showed high similarity among the five sequences. Such similarity is consistent with the sequences belonging to the same broad beta-lactamase gene family. Nevertheless, the exact degree of similarity must be verified because the current matrix contains an unresolved inconsistency involving the two sequences labelled TEM-116.

2. Nucleotide variation

Eight variable nucleotide positions were reported in the multiple sequence alignment. Nucleotide substitutions can be synonymous or nonsynonymous, depending on whether they alter the encoded amino acid.

The presence of nucleotide differences alone does not establish that an enzyme's activity, stability or antibiotic substrate profile has changed. Those interpretations require a validated coding sequence and appropriate functional or literature evidence.

3. Phylogenetic relationships

The phylogenetic tree provides a visual representation of the relationships inferred from the selected sequences. Sequences that group together are more similar under the selected analysis, but the strength of each inferred grouping depends on the method and branch-support values.

Because only five sequences were analyzed, the tree provides a limited comparison of this selected set. It should not be interpreted as a comprehensive evolutionary history of all TEM-type beta-lactamases or of the bacterial species from which they were obtained.

4. Protein-level variation

The initial protein comparison reported five variable amino acid positions. If these differences are confirmed using correctly translated coding sequences, they may help distinguish the encoded TEM variants.

However, amino acid substitutions cannot be assumed to increase or decrease beta-lactamase activity without supporting experimental data or reliable functional evidence. The initial translation also requires validation before the reported protein lengths and variation counts can be finalized.

5. Importance of bioinformatics in antibiotic resistance research

This project demonstrates how publicly available nucleotide data and computational tools can be combined to investigate genetic variation associated with antibiotic resistance.

Sequence alignment, pairwise identity calculations, phylogenetic analysis and protein translation provide complementary forms of evidence. Together, these approaches help organize and compare sequence data and generate hypotheses for further study.

6. Limitations

The study has several limitations:

1. Only five selected sequences were included.
2. The preliminary pairwise identity results require verification against the final alignment.
3. The coding frame and sequence boundaries must be confirmed before finalizing protein-level conclusions.
4. Phylogenetic interpretation depends on the tree-building method and availability of branch-support values.
5. Sequence similarity alone cannot establish antibiotic susceptibility, clinical significance or enzyme activity.
6. No laboratory experiments were performed to measure beta-lactamase activity or antibiotic resistance.

CONCLUSION

This project applied bioinformatics methods to the comparative analysis of five selected TEM-type beta-lactamase gene sequences. The workflow included multiple sequence alignment, pairwise nucleotide identity analysis, phylogenetic tree construction, identification of variable nucleotide positions and preliminary protein sequence comparison.

The initial analysis reported an alignment length of 861 nucleotide positions, eight variable nucleotide positions and five variable amino acid positions. The phylogenetic tree provided a means of visualizing relationships among the selected sequences.

The study demonstrates the usefulness of bioinformatics in comparing antibiotic-resistance-associated genes and identifying sequence differences.

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